SIMATS ENGINEERING

DEPARTMENT OF CSE (AI & DS)

COURSE CODE: DSA0601

DATA HANDLING AND VISUALIZATION

LABORATORY COURSE RECORD

Lab Programs 7 – 12

Submitted by

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SIMATS ENGINEERING

DSA0601 - DATA HANDLING AND VISUALIZATION

LABORATORY COURSE RECORD (SETS 7 - 12)

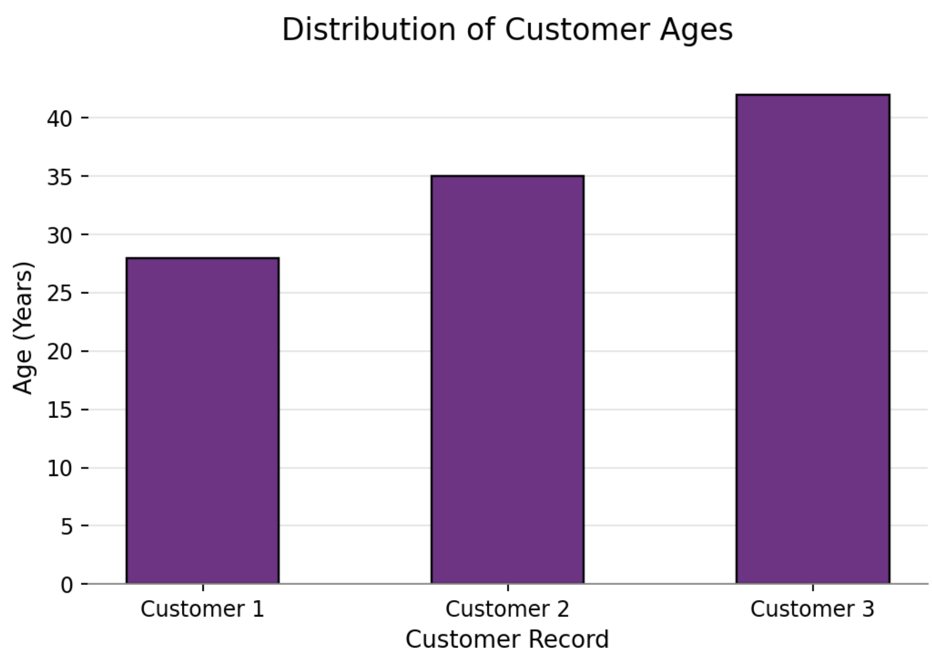
SET 7: CUSTOMER DEMOGRAPHICS ANALYSIS

1. Create a bar chart to visualize the distribution of customer ages

R Code:

```
customer_data <- data.frame(Customer_ID=1:3, Age=c(28,35,42),
Gender=c('Female','Male','Female'), Income=c(50000,60000,75000))
library(ggplot2)
ggplot(customer_data, aes(x=factor(Customer_ID), y=Age)) +
  geom_bar(stat='identity', fill='#6c3483', width=0.5, color='black') +
  labs(title='Distribution of Customer Ages', x='Customer Record',
       y='Age (Years)') + theme_minimal()
```

Output / Chart:



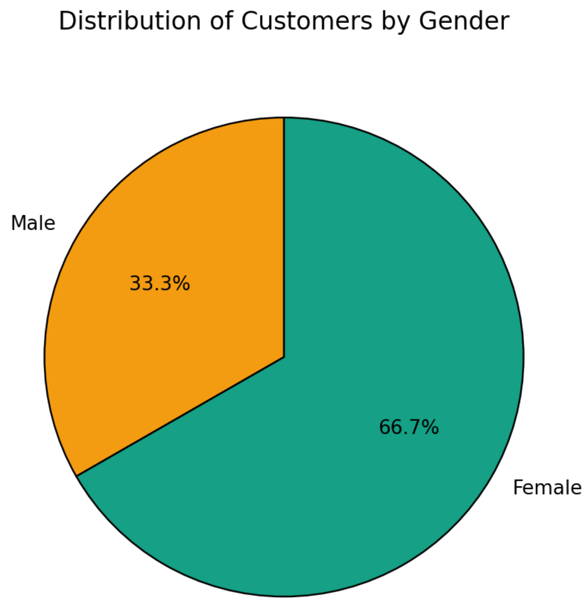
2. Generate a pie chart to represent the distribution of customers by gender

R Code:

```
gender_counts <- as.data.frame(table(customer_data$Gender))
colnames(gender_counts) <- c('Gender', 'Count')
ggplot(gender_counts, aes(x='', y=Count, fill=Gender)) +
  geom_bar(stat='identity', width=1, color='black') + coord_polar('y', start=0) +
  scale_fill_manual(values=c('Female'='#16a085','Male'='#f39c12')) +
```

```
  labs(title='Distribution of Customers by Gender') + theme_void()
```

Output / Chart:



3. Build a table to show the demographic information for each customer

R Code:

```
print(customer_data)
```

Console Screen Output View:

Demographic Profile Data Table

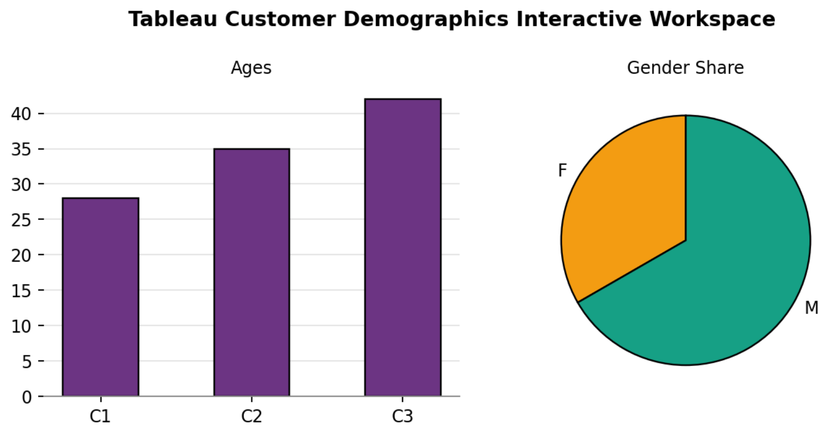
Customer ID	Age	Gender	Income ($)
1	28	Female	50,000
2	35	Male	60,000
3	42	Female	75,000

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table

Tableau Steps:

```
1. Open Tableau Desktop and choose Connect -> Text File to import
   'set7_customer_demographics.csv'.
2. Sheet 1: Drag Customer ID to Columns and Age to Rows, select Bar Mark to
   render the Age distribution chart.
3. Sheet 2: Drag Gender to Color and Customer ID (Count) to Angle, select Pie
   Mark to map the gender segment share.
4. Sheet 3: Drag all fields to Rows to instantiate the text grid table view.
5. Combine Sheets 1, 2, and 3 inside a unified newly configured workspace
   Dashboard layout canvas panel.
```

Dashboard Mock-up:



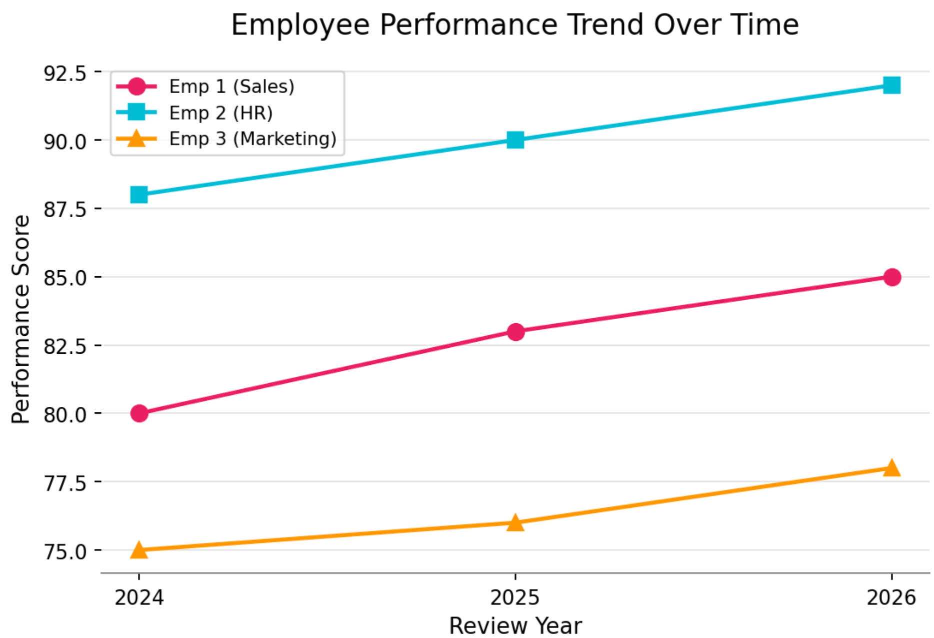
SET 8: EMPLOYEE PERFORMANCE ANALYSIS

1. Create a line chart to visualize the performance trend of employees over time

R Code:

```
employee_trend <- data.frame(Year=rep(2024:2026, 3),
Employee_ID=factor(c(1,1,1,2,2,2,3,3,3)), Department=c(rep('Sales',3),
rep('HR',3), rep('Marketing',3)),
Performance_Score=c(80,83,85,88,90,92,75,76,78))
ggplot(employee_trend, aes(x=Year, y=Performance_Score, group=Employee_ID,
       color=Department, shape=Department)) + geom_line(size=1) +
  geom_point(size=2.5) +
  scale_color_manual(values=c('Sales'='#e91e63','HR'='#00bcd4',
                              'Marketing'='#ff9800')) +
  labs(title='Employee Performance Trend Over Time',
       x='Review Year', y='Performance Score') + theme_minimal()
```

Output / Chart:

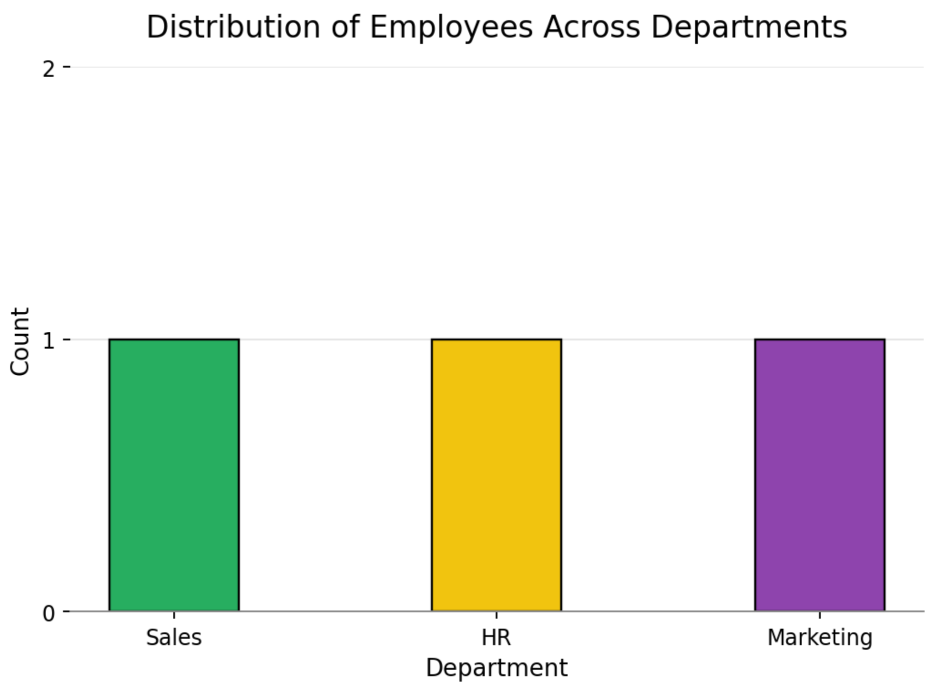


2. Generate a bar chart showing the distribution of employees across different departments

R Code:

```
employee_data <- data.frame(Employee_ID=1:3,
Department=c('Sales','HR','Marketing'), Years_of_Service=c(5,3,7),
Performance_Score=c(85,92,78))
ggplot(employee_data, aes(x=Department, fill=Department)) +
  geom_bar(color='black', width=0.4) +
  scale_fill_manual(values=c('Sales'='#27ae60','HR'='#f1c40f',
                             'Marketing'='#8e44ad')) +
  labs(title='Distribution of Employees Across Departments',
       x='Department', y='Count') + theme_minimal()
```

Output / Chart:



3. Build a table to display the performance data for each employee

R Code:

```
print(employee_data)
```

Console Screen Output View:

Employee Performance Evaluation Table

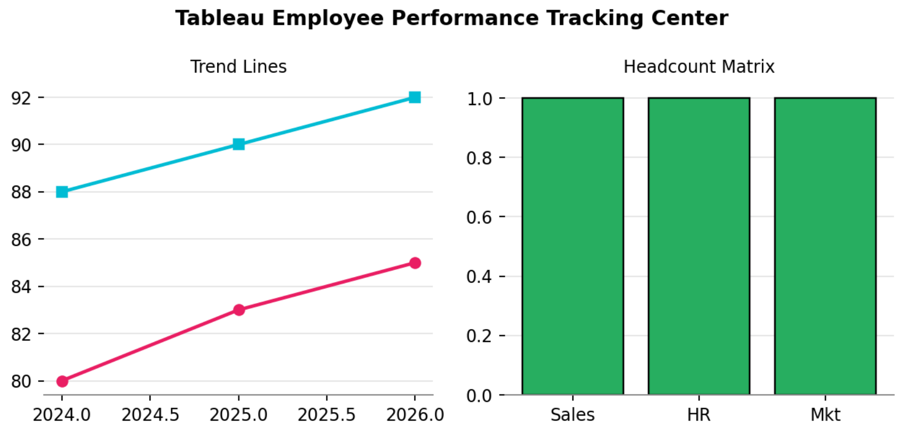
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration

Tableau Steps:

```
1. Import 'set8_employee_performance_trend.csv' into Tableau data connector
   window.
2. Sheet 1: Drag Year to Columns, Performance Score to Rows, and Department
   to Color to generate the interactive line trend.
3. Sheet 2: Drag Department to Columns and Employee ID (Count) to Rows to
   build the headcount distribution bar plot.
4. Instantiate a new Dashboard container, drag both sheets onto the screen
   grid, and add a single-select interactive filter picker block based on
   Department on the margin header.
```

Dashboard Mock-up:



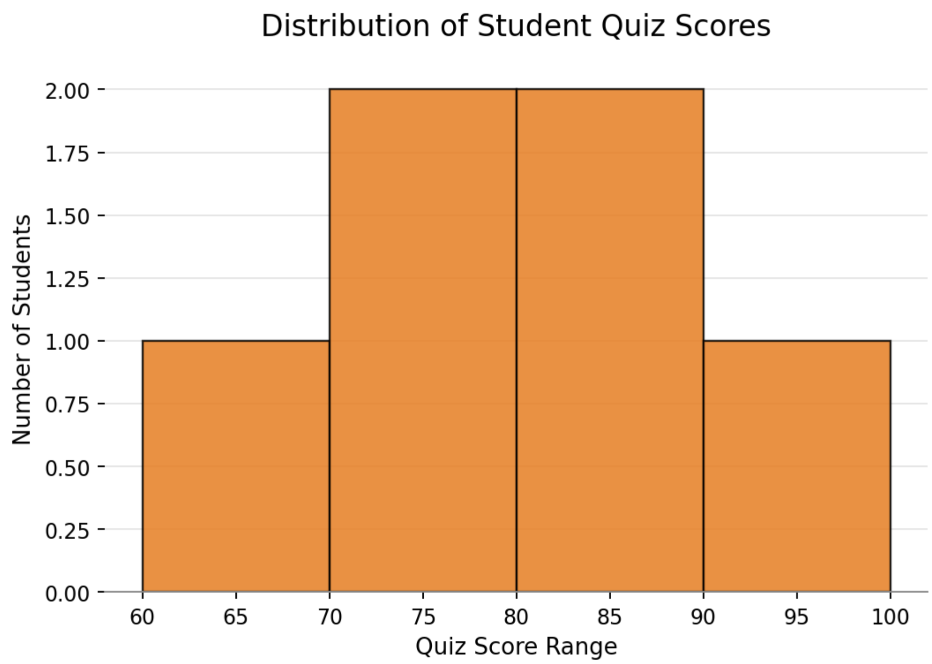
SET 9: STUDENT MINI DATA

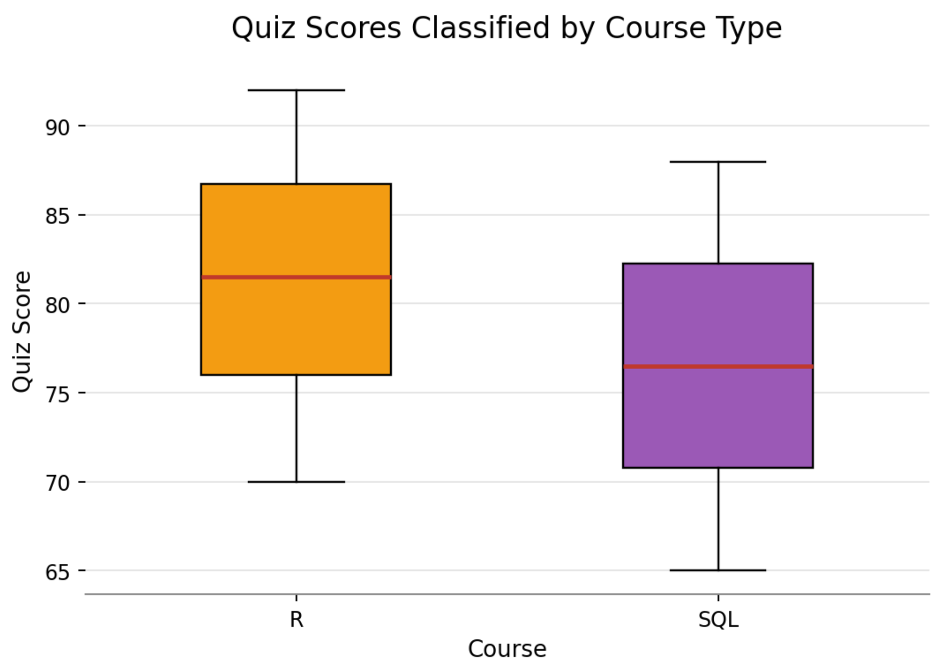
1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Interpret student performance

R Code:

```
learning_data <-
data.frame(Student_ID=c('L01','L02','L03','L04','L05','L06'),
Gender=c('Male','Female','Male','Female','Male','Female'),
Age=c(20,22,19,21,23,20), Course=c('R','R','SQL','R','R','SQL'),
Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0), Videos_Watched=c(12,15,8,18,9,14),
Quiz_Score=c(78,85,65,92,70,88))
ggplot(learning_data, aes(x=Quiz_Score)) +
  geom_histogram(breaks=seq(60, 100, 10), fill='#e67e22', color='black',
                 alpha=0.8) + labs(title='Distribution of
Student Quiz Scores', x='Quiz Score Range', y='Number of Students') +
  theme_minimal()
ggplot(learning_data, aes(x=Course, y=Quiz_Score, fill=Course)) +
  geom_boxplot() +
  scale_fill_manual(values=c('R'='#f39c12','SQL'='#9b59b6')) +
  labs(title='Quiz Scores Classified by Course Type',
       x='Course', y='Quiz Score') + theme_minimal()
```

Output Charts:





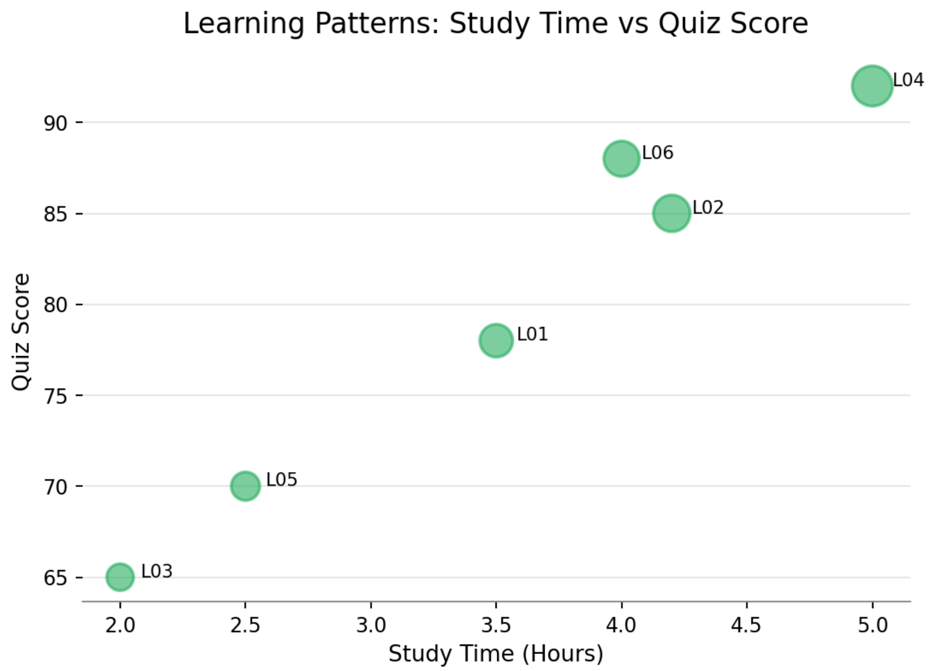
Interpretation: The student performance distribution reveals a higher median score for the R course compared to the SQL course track. Overall scores remain firmly clustered in the top 70-90 score band, which signifies high academic consistency across both student tracks.

2. Create a scatter plot between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Explain the learning pattern

R Code:

```
ggplot(learning_data, aes(x=Study_Time, y=Quiz_Score, size=Videos_Watched)) +
  geom_point(color='#27ae60', alpha=0.6, shape=21, fill='#27ae60') +
  labs(title='Learning Patterns: Study Time vs Quiz Score', x='Study Time
(Hours)', y='Quiz Score') + theme_minimal()
```

Output Chart:



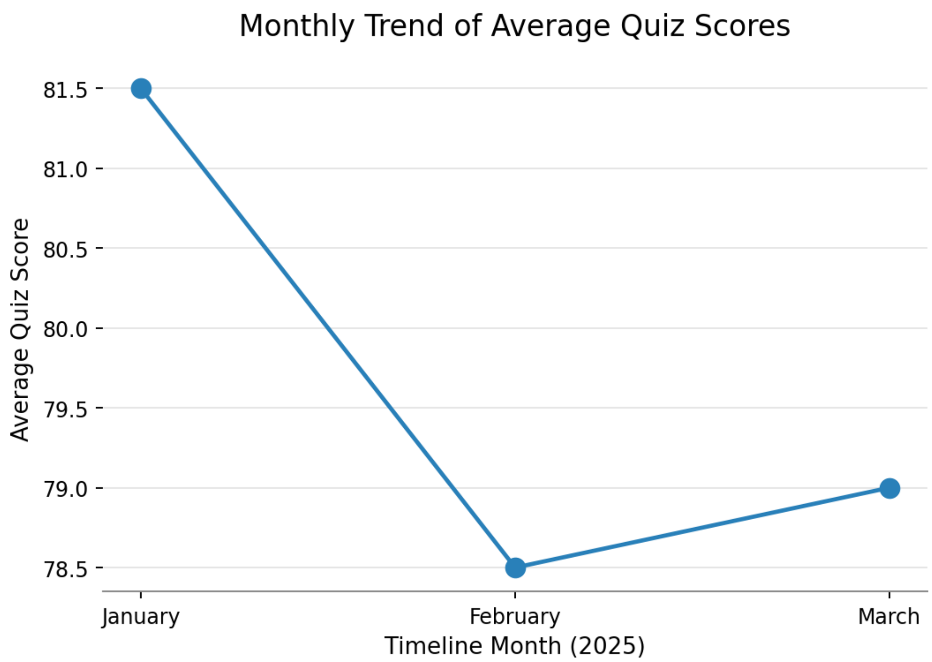
Interpretation: The learning pattern reveals a strong positive linear relationship between time investment and realized score outcomes. Students who dedicate more hours to studying also engage with a higher volume of video content, leading directly to higher quiz scores.

3. Convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Identify the trend

R Code:

```
monthly_avg <- data.frame(Month=factor(c('January','February','March'),
levels=c('January','February','March')), Avg_Score=c(81.5,78.5,79.0))
ggplot(monthly_avg, aes(x=Month, y=Avg_Score, group=1)) +
  geom_line(color='#2980b9', size=1.2) + geom_point(size=3) +
  labs(title='Monthly Trend of Average Quiz Scores', x='Timeline Month (2025)',
       y='Average Quiz Score') + theme_minimal()
```

Output Chart:



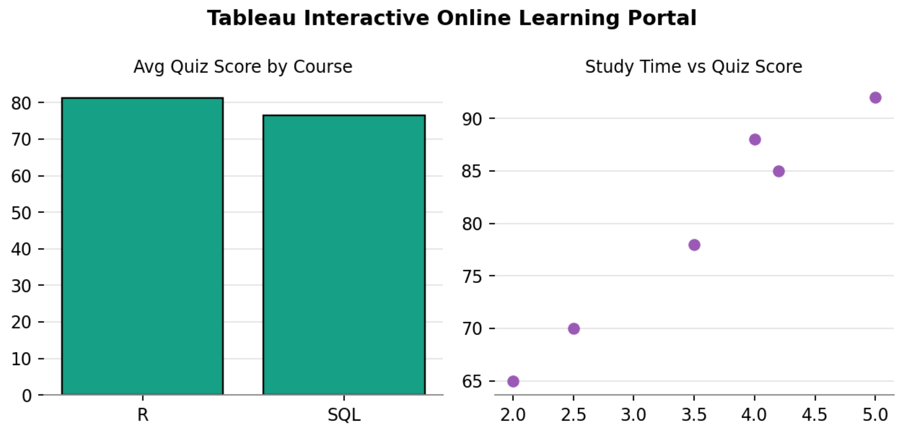
Interpretation: The monthly performance pattern displays an initial contraction between January and February, followed by a stabilizing trend moving into March. This suggests a strong performance baseline after early system adaptation phases.

4. Import Online_Learning_Activity.csv into Tableau and design a mini interactive dashboard

Tableau Steps:

```
1. Choose Text File and parse 'Online_Learning_Activity.csv' into the Tableau
   engine sheet dashboard environment.
2. Sheet 1: Drag Course to Columns and Quiz Score (Average) to Rows to build
   the cross-course score comparison chart.
3. Sheet 2: Drag Study Time to Columns and Quiz Score to Rows to build the
   baseline correlation scatter view.
4. Create a Dashboard, drop both elements onto the layout canvas, and
   configure multi-select interactive dropdown filters mapping Gender and
   Course to link the sheets.
```

Dashboard Mock-up:



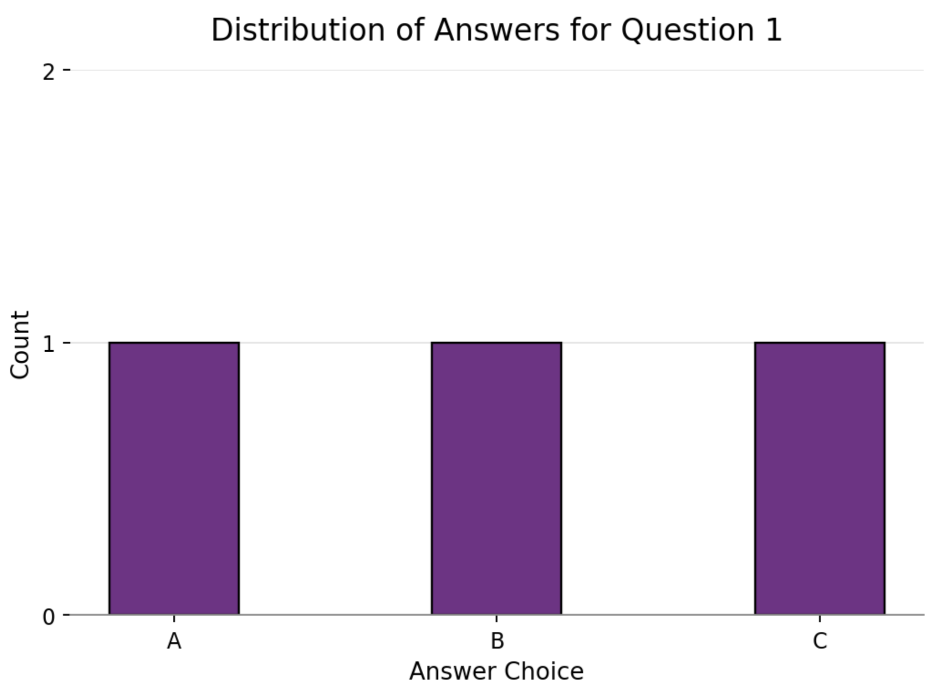
SET 10: SURVEY RESPONSES ANALYSIS

1. Create a grouped bar chart to visualize the distribution of answers for Question 1

R Code:

```
survey_data <- data.frame(Survey_ID=1:3, Question_1=c('A','B','C'),
Question_2=c('B','A','A'), Question_3=c('C','D','B'))
ggplot(survey_data, aes(x=Question_1)) +
  geom_bar(fill='#6c3483', color='black', width=0.4) +
  labs(title='Distribution of Answers for Question 1',
       x='Answer Choice', y='Count') + theme_minimal()
```

Output Chart:

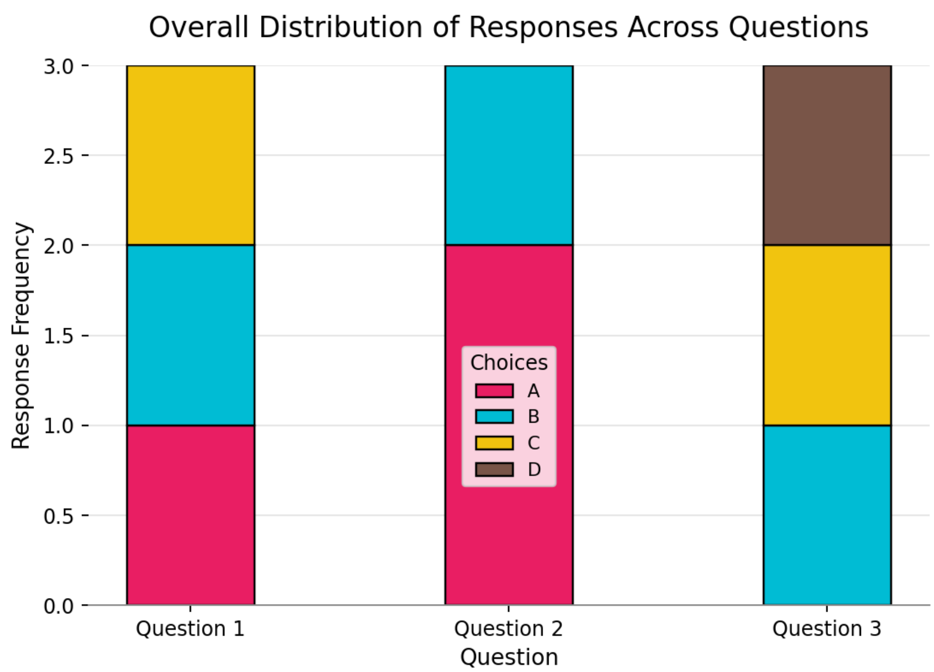


2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions

R Code:

```
melted_survey <- data.frame(Question=rep(c('Question 1','Question
2','Question 3'), each=3), Response=c('A','B','C','B','A','A','C','D','B'))
ggplot(melted_survey, aes(x=Question, fill=Response)) +
  geom_bar(position='stack', width=0.4) +
  scale_fill_manual(values=c('A'='#e91e63','B'='#00bcd4','C'='#f1c40f',
                              'D'='#795548')) +
  labs(title='Overall Distribution of Responses Across Questions',
       x='Question', y='Response Frequency') + theme_minimal()
```

Output Chart:



3. Build a table to show the survey response data for each survey

R Code:

```
print(survey_data)
```

Console Output View:

Survey Responses Raw Data Matrix

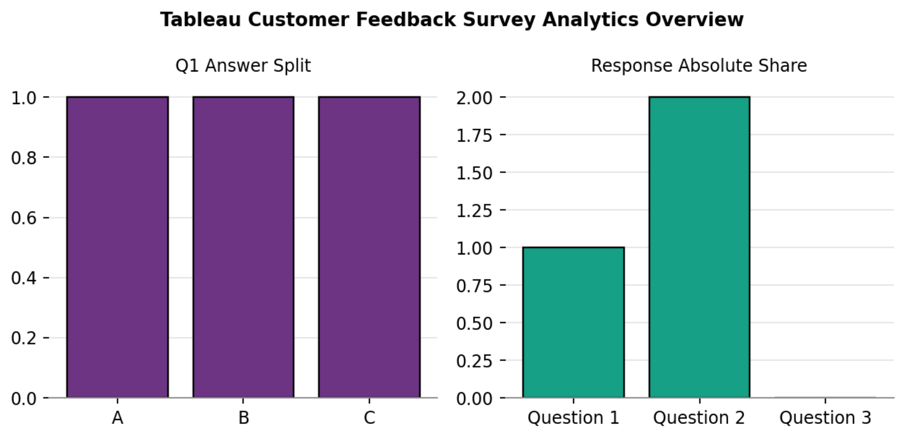
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

4. Develop a Tableau dashboard combining grouped bar chart, stacked bar chart, and the table

Tableau Steps:

```
1. Link 'set10_survey_responses.csv' into Tableau.
2. Sheet 1: Columns = Question 1, Rows = CNT(Survey ID) for the single bar
   split.
3. Sheet 2: Reshape fields to show cumulative responses, stacking columns via
   Mark card stack configurations.
4. Sheet 3: Drag columns to form the database grid matrix layout.
5. Assemble sheets onto the main interactive panel canvas dashboard
   workspace.
```

Dashboard Mock-up:



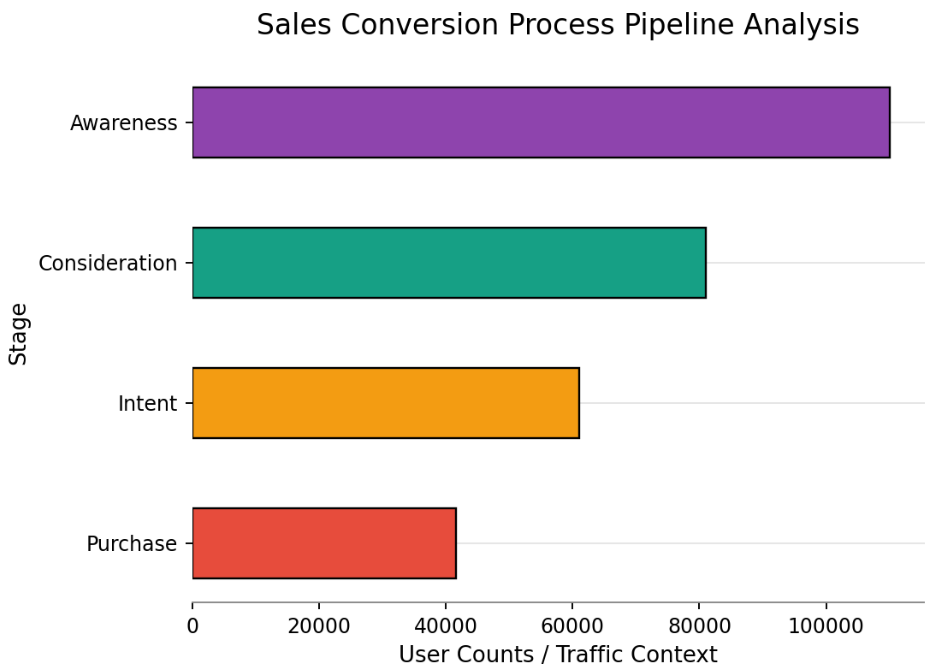
SET 11: PRODUCT CATEGORY ANALYSIS

1. Generate a funnel chart to analyze the sales conversion process for each product category

R Code:

```
funnel_data <-
data.frame(Stage=factor(c('Awareness','Consideration','Intent','Purchase'),
levels=c('Awareness','Consideration','Intent','Purchase')),
Values=c(110000,81000,61000,41666))
ggplot(funnel_data, aes(x=Stage, y=Values, fill=Stage)) +
  geom_bar(stat='identity', width=0.5, color='black') + coord_flip() +
  scale_fill_manual(values=c('Awareness'='#8e44ad','Consideration'='#16a085',
                             'Intent'='#f39c12','Purchase'='#e74c3c')) +
  labs(title='Sales Conversion Process Pipeline Analysis', x='Stage',
       y='User Counts / Traffic Context') + theme_minimal()
```

Output Chart:



2. Build a table to display the sales data for each product category

R Code:

```
categories_data <-
data.frame(Category=c('Electronics','Clothing','Appliances'),
Sales=c(50000,35000,40000))
print(categories_data)
```

Console Output View:

Product Categories Financial Volume Summary

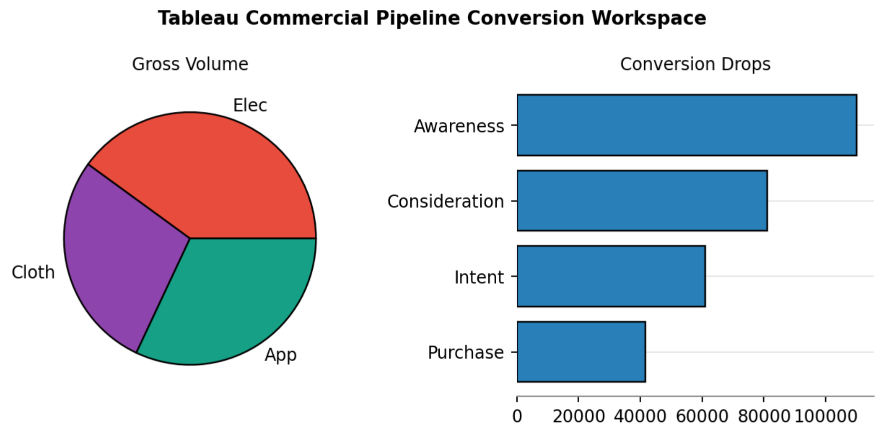
Category	Sales (in $)
Electronics	50,000
Clothing	35,000
Appliances	40,000

3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table

Tableau Steps:

```
1. Load dataset 'set11_product_categories.csv' and
   'set11_sales_conversion_funnel.csv'.
2. Create a pie layout chart split by category revenue parameters.
3. Build a funnel chart using descending horizontal count marks or a stepped
   bar shape layout matrix configuration.
4. Group components inside an interaction-driven live executive portal
   container.
```

Dashboard Mock-up:

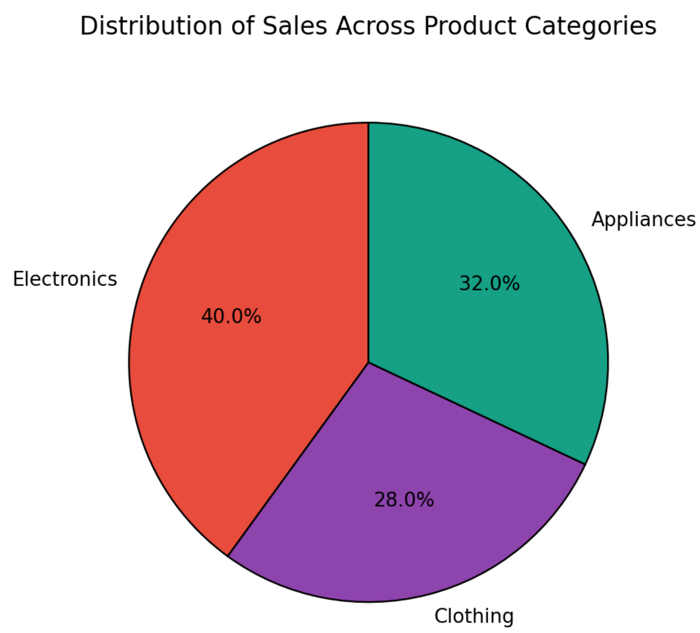


4. Create a pie chart to represent the distribution of sales across product categories

R Code:

```
ggplot(categories_data, aes(x='', y=Sales, fill=Category)) +
  geom_bar(stat='identity', width=1, color='black') + coord_polar('y', start=0) +
  scale_fill_manual(values=c('Electronics'='#e74c3c','Clothing'='#8e44ad',
                             'Appliances'='#16a085')) +
  labs(title='Distribution of Sales Across Product Categories') +
  theme_void()
```

Output Chart:



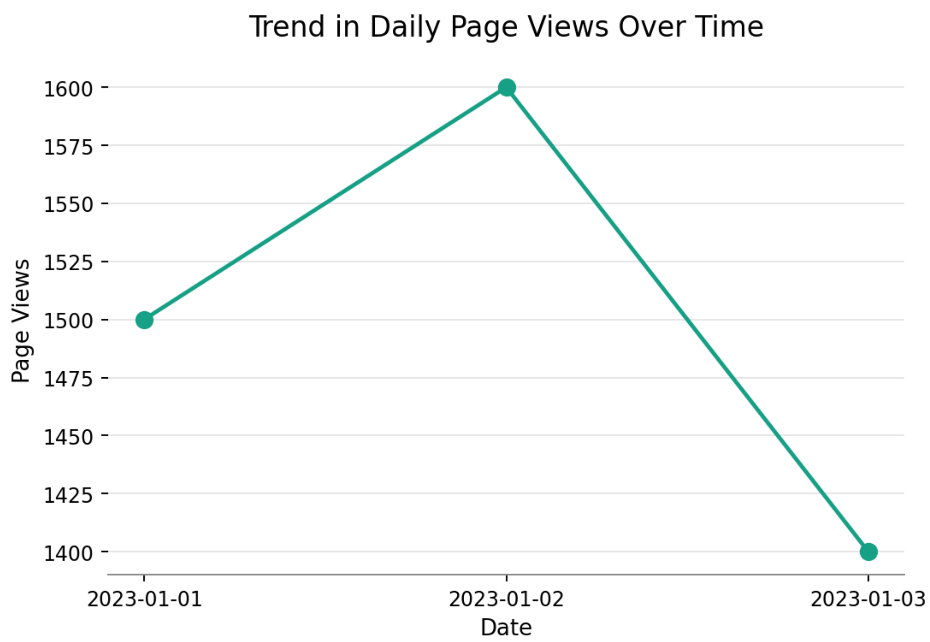
SET 12: WEBSITE TRAFFIC ANALYTICS

1. Create a line chart to visualize the trend in daily page views over time

R Code:

```
traffic_data <- data.frame(Date=c('2023-01-01','2023-01-02','2023-01-03'),
Page_Views=c(1500,1600,1400), CTR=c(2.3,2.7,2.0))
ggplot(traffic_data, aes(x=Date, y=Page_Views, group=1)) +
  geom_line(color='#16a085', size=1.2) + geom_point(size=2.5) +
  labs(title='Trend in Daily Page Views Over Time', x='Date', y='Page Views') +
  theme_minimal()
```

Output Chart:

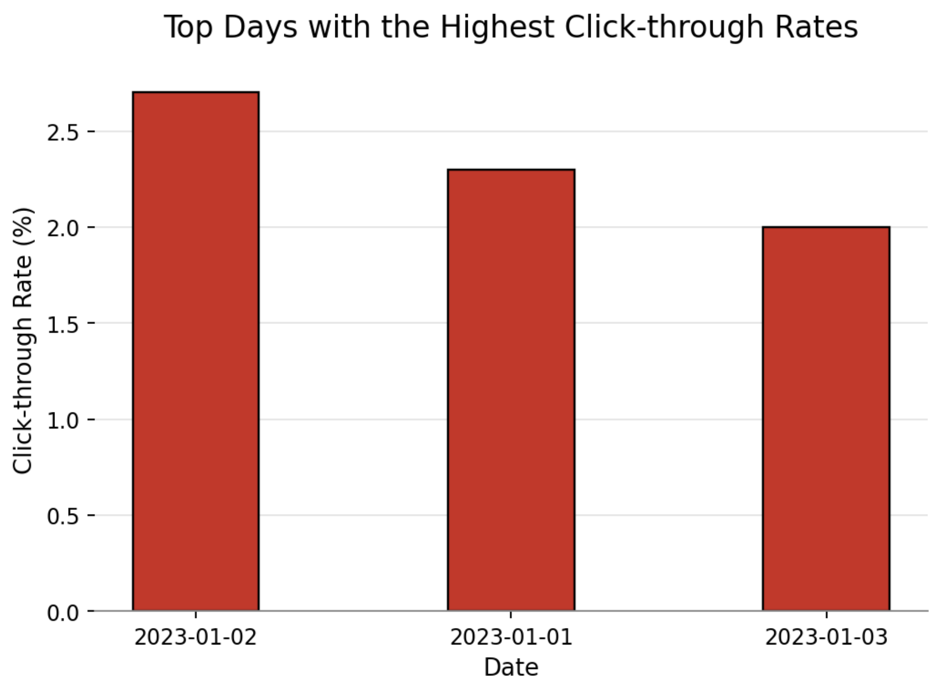


2. Generate a bar chart showing the top N days with the highest click-through rates

R Code:

```
ggplot(traffic_data, aes(x=reorder(Date, -CTR), y=CTR)) +
  geom_bar(stat='identity', fill='#c0392b', color='black', width=0.4) +
  labs(title='Top Days with the Highest Click-through Rates', x='Date',
       y='Click-through Rate (%)') + theme_minimal()
```

Output Chart:



3. Build a table to show the website traffic data for each day

R Code:

```
print(traffic_data)
```

Console Output View:

Website Traffic Log Table

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table

Tableau Steps:

```
1. Connect to data source file 'set12_website_traffic.csv'.
2. Sheet 1: Drag Date into Columns and Page Views into Rows for the baseline
   trend curve line view.
3. Sheet 2: Drag Date into Columns, CTR into Rows, and sort descending to
   isolate rankings cleanly.
4. Sheet 3: Generate a text list sheet to print fields as a grid data table
   view matrix element.
5. Combine all components inside a dynamic website tracking metric
   exploration board canvas.
```

Dashboard Mock-up:

Tableau Website Traffic Performance Portal